Reinforcement Learning

Tools for sequential decision making under uncertainty

Fathers of the Deep Learning Revolution Receive ACM A.M. Turing Award

https://awards.acm.org/about/2018-turing

Yoshua Bengio

Geoffrey Hinton

Yann LeCun

They Trained Artificial Neural Networks Using Physics

https://www.nobelprize.org/prizes/physics/2024/press-release/

John Hopfield

Geoffrey Hinton

They Cracked the Code for Proteins' Amazing Structures

https://www.nobelprize.org/prizes/chemistry/2024/press-release/

David Baker **Demis Hassabis** **John Jumper**

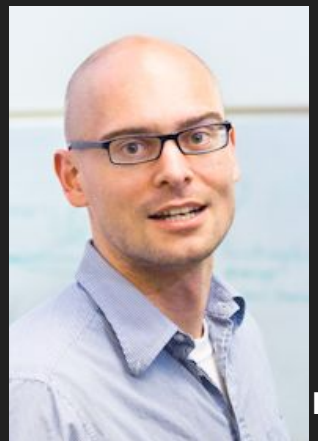
A Few Well-known Faces

Andrew Ng

Ilya Sutskever

David Silver

John Schulman



Pieter Abbeel

Reinforcement Learning (RL)

RL is machine learning (ML) in which the learning system *adjusts* its *behavior* to *maximize* the *amount of reward* and *minimize* the *amount of punishment* it receives *over time* while *interacting* with its *environment*.

- *Rediscovering Reinforcement Learning*
- Andrew G. Barto, CACM, Oct 23 2025
- https://cacm.acm.org/federal-funding-of-academic-research/rediscovering-reinforcement-learning/

The RL Configuration

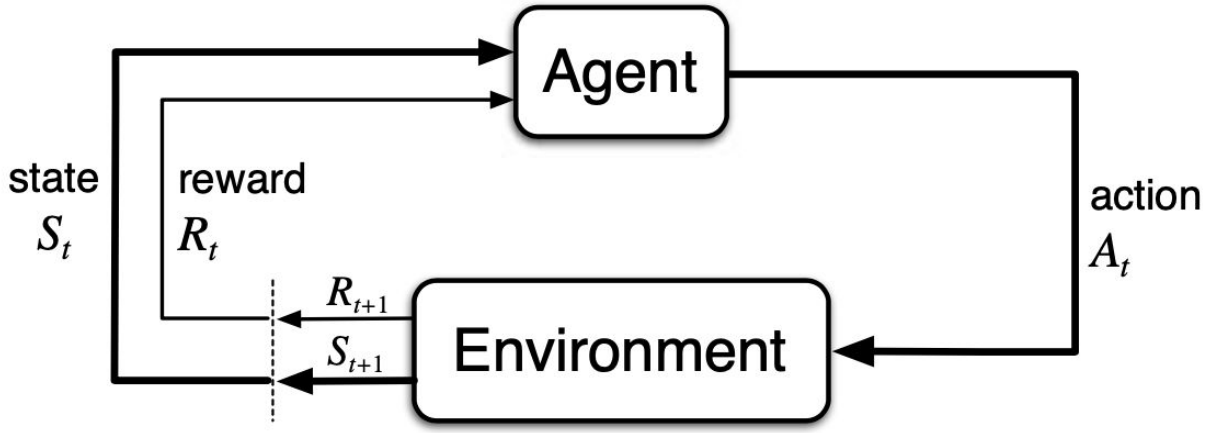


Figure 3.1: The agent–environment interaction in a Markov decision process.

Agent is the learner and the decision maker
Agent only performs actions
Environment rewards actions and indicates the new state

Primitives

- Cause and effect
- Actions and their consequences
- Learning by trial-and-error
  - *without a teacher*
- Learning from interactions

Meaning of the word *Reinforcement Learning*

The word means three different ideas

- The problem of RL
- The class of solution methods that work well on the problem
- The field that studies the problem and its solution methods

Meaning of the word *Reinforcement Learning*

The word means three different ideas

- The problem of RL
  - Incompletely known dynamical systems - Markov decision processes
  - Sensation, action, and goal
  - State, action, and goals relating to the states of the environment
- The class of solution methods that work well on the problem
  -
- The field that studies the problem and its solution methods
  - Different from supervised learning
  - Different from unsupervised learning

Supervised Learning (SL)

- SL relies on carefully *labeled* data sets that provide ground truth in order for an AI system to gain an understanding of the world and perform their assigned tasks.

- An agent learns by passively observing example input/output pairs provided by a “teacher.”

Supervised Learning (SL)

- SL relies on carefully *labeled* data sets that provide ground truth in order for an AI system to gain an understanding of the world and perform their assigned tasks.
- An agent learns by passively observing example input/output pairs provided by a “teacher.”
- SL works to reduce errors between responses and correct responses as given by training examples.
- SL works great in relatively well-circumscribed domains for which one can collect large amounts of labeled data, and for which the type of inputs seen during deployment are not too different from the ones used during training.

Differences Between RL and SL

- RL does not rely on knowledge of correct responses. It relies on more general evaluations of behavior.
- Aims to build AI systems that will be able to learn from the world with minimal help from humans.

Learning without Supervision

By contrast, animals and humans seem to learn complex tasks from a very small amount of interactions with the real world. One hypothesis is that animals and humans learn vast amounts of knowledge about how the world works through mere observation and occasional actions.

Learning without Supervision

Good predictive world models are an essential component of intelligent behavior: with them, one can predict outcomes and plan courses of actions.

- *one could argue that prediction is the essence of intelligence*.

Good predictive models may be the basis of intuition, intuitive physics, reasoning and "*common sense*", allowing us to fill in missing information:

- *predicting the future from the past and present*
- *predicting* the state of the world from noisy percepts.

RL is not Unsupervised Learning

- Unsupervised learning tries to find structures hidden on collection of unlabeled data
- RL tries to maximize a reward signal, that’s all.
- RL does not deal with unravelling the hidden structures.

The distinguishing challenge in RL is the trade-off between *exploration* and *exploitation*.

- The agent must try a variety of actions and adopt those that appear to be best.

Agent, Goals and interactions with Uncertain Environment

- RL explicitly considers the whole problem of a goal-directed agent interacting with a stochastic environment for maximizing rewards.
- Planning and real-time decision making
- Predictive models for planning

- The effect of actions cannot be fully predicted.
- The agent must monitor the environment frequently and react appropriately.

Learning without Supervision

- *AI Doesn't Need Our Supervision*
- IEEE Spectrum, 22 Feb 2022
- https://spectrum.ieee.org/yann-lecun-ai

"Deep learning is causing revolutions in computer perception and natural language understanding. But almost all these successes largely rely on supervised learning, where the machine is trained with a large number of human-annotated samples.

For control and game AI, most systems use model-free reinforcement learning, which requires too many trials to be practical in the real world."

Elements of RL

- Policy
- Reward Signal
- Value Function

- Model of the Environment (optional)

Elements of RL

- Policy
  - Behaviors in a state.
  - Choosing from a set of actions in a state.
  - May involve computationally extensive search or a cheap table lookup
  - May be stochastic, specifying probabilities for each action

Elements of RL

- Policy
- Reward Signal
  - A single number supplied by the environment
  - immediate intrinsic desirability of environment states
  - Agent tries to maximize the total reward it receives over a long run
  - Reward signal is the primary basis for altering the policy
  - Reward signals may be stochastic functions of the state of the environment and the actions taken
- Value Function

Elements of RL

- Policy
- Reward Signal
- Value Function
  - Specifies what is good in the long run
  - Value of a state is the total reward the agent can expect to accumulate in the future starting from that state
  - Long-term desirability of states
  - Takes into account states that are likely to follow and rewards available in those states
- A state may *always* yield low reward and yet may have high value
  - It is regularly followed by other states that yield high rewards
- The opposite may also be true!
- Model of the Environment (optional)

Greatest Value versus Rewards

Value Function

- Without rewards there could be no values
- The sole purpose estimating values is to achieve more reward
- Action choices are made based on value judgements
- An agent seeks highest value, not highest reward
- Rewards are given directly by the environment
- Values must be estimated and re-estimated from the sequence of observations an agent makes over its entire lifetime

RL Setup for the Agent

RL can be formalized in terms of **Markov decision processes** (MDPs), in which the agent initially only knows

- the set of possible states and
- the set of possible actions.

The dynamics, $P(s'|a,s)$, and the reward function, $R(s,a)$, are not given to the agent.

As in an MDP, after each action, the agent observes the state it is in and receives a reward.